

Zach Taylor

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SKILLS

Languages: C++, Python, Golang, Assembly, Typescript, JavaScript, HTML, CSS, SQL, Verilog
Libraries/Frameworks: MFC, React, Node.js, Next.js, Tailwind, TensorFlow, CUDA, FastAPI, MySQL
Developer Tools: Git, Docker, Linux, Visual Studio, Jenkins, Google Test, AWS, Vercel, SIMD, Microcontrollers

EXPERIENCE

Software Engineer Intern *Rohde & Schwarz* April 2024 – Jan 2025

- Enabled advanced **network analysis** for the U.S. Army, Tesla, and first responders by delivering custom scanning features and frequency band configurations designed, implemented, and integrated using **C++**.
- Achieved a **25%** improvement in **processing speed** and maintainability by redesigning programs with efficient lookup algorithms and modern library features.
- Ensured stability and reliability across multiple software components by uncovering edge cases through rigorous **unit testing** and debugging with **Google Test**.
- Led meetings and cross-functional collaboration to support team alignment and ensure compliance with **agile methodologies**.

Software Development Intern *Portland State University* June 2019 – September 2019

- Achieved **97%** accuracy in object detection by implementing a model with **Python OpenCV** for image capture and leveraging **TensorFlow's** transfer learning APIs.
- Delivered a versatile image processing solution by configuring the model to handle both live camera feeds and static image datasets for real time and batch applications.
- Directed team planning** and streamlined project execution by managing weekly deliverables and fostering innovative problem solving through open collaboration.

Undergraduate Teaching Assistant *Oregon State University* September 2020 – June 2023

- Strengthened students' foundational and practical skills by teaching **data structures** and **operating systems** in C, and guiding full stack web development using **JavaScript**, **Node.js**, and **HTML/CSS**.
- Raised academic performance by **30%** among struggling students by delivering personalized one-on-one and small group tutoring focused on reinforcing core concepts and addressing individual challenges.
- Improved students' **problem solving** abilities and debugging proficiency by walking through real code issues and encouraging a deeper understanding of troubleshooting techniques.

PROJECTS

Patient Monitoring Robot | *Python, OpenCV, ROS2* [GitHub](#)

- Enhanced safety in nursing homes by achieving **96%** accuracy in detecting falls and abnormal movements through the development of a robotic monitoring system using **Python**, **OpenCV**, and **Linux**.
- Boosted system performance by **56%** and enabled seamless camera integration by overhauling a legacy ROS2 codebase to meet the latest framework standards.
- Verified software performance and accuracy across various settings through comprehensive field tests, utilizing real-time motion tracking to replicate typical user environments and edge cases.
- Drove team progress** across the development lifecycle by facilitating technical discussions, aligning objectives, removing blockers, and ensuring clear communication.

SIMD Image Filter | *C++, SIMD* [Full Site](#)

- Designed and implemented a high performance image filtering library in **C++**, featuring blur, sharpen, and edge detection algorithms.
- Optimized image processing routines using **SIMD intrinsics**, resulting in **5x** faster execution compared to scalar implementations.

EDUCATION

Oregon State University September 2019 – June 2023
Bachelor of Science in Computer Science *Corvallis, OR*